

AWS Infrastructure Cost Analysis

MERN Stack Video Education Platform

Cost Model - AWS

The following assumptions were used when calculating the infrastructure cost for the platform.

Platform Architecture

The system uses a **MERN stack architecture**:

- **MongoDB** – database layer
- **Express.js** – backend framework
- **React** – frontend application
- **Node.js** – backend runtime

Infrastructure components include:

- Amazon S3 – video storage
 - Amazon CloudFront – global CDN delivery
 - Amazon EC2 – backend API servers
 - Application Load Balancer – traffic distribution
 - Amazon CloudWatch – monitoring
 - Amazon Route 53 – DNS management
 - MongoDB Atlas – managed database
-

User Behaviour Assumption

To estimate bandwidth usage, the following usage pattern was assumed:

Every user watches every video once per month.

This represents a **worst-case bandwidth scenario** and provides a safe upper bound for infrastructure planning.

Content Library Scenarios

Two content library sizes were evaluated:

Library Size	Total Video Storage
50 videos	8.32 GB
100 videos	16.64 GB

User Scale Scenarios

Four user growth scenarios were evaluated:

Users

500

5,000

50,000

500,000

1. Amazon S3 Storage Cost

Purpose

Amazon S3 stores all platform media assets including:

- course videos
- video thumbnails
- supporting learning resources
- platform static assets

S3 provides **99.999999999% durability**, making it suitable for storing long-term educational content.

Storage Requirements

50 Video Library

Videos	Storage
50	8.32 GB

100 Video Library

Videos	Storage
100	16.64 GB

Monthly Storage Cost

Scenario	Monthly Cost
50 videos	\$3.46
100 videos	\$6.92

Observation

Storage cost is **extremely small compared to video streaming bandwidth**.

2. CloudFront CDN Bandwidth Cost

Purpose

Amazon CloudFront is responsible for delivering video content globally.

Instead of users downloading videos directly from S3, CloudFront:

- caches content at global edge locations
- reduces latency for international users
- reduces load on origin infrastructure
- improves streaming performance

Example Bandwidth Usage

For **500 users watching 50 videos**:

Metric	Value
Total Video Views	25,000
Total Bandwidth	~1 TB (approx.)

Bandwidth scales linearly with the number of users.

Monthly CloudFront Cost — 50 Video Library

Scenario 1 — No Bandwidth Protection

Users	CloudFront Cost
500	\$106
5,000	\$634
50,000	\$5,926
500,000	\$58,846

Scenario 2 — Signed URL Protection (30% Reduction)

Users	CloudFront Cost
500	\$74.20
5,000	\$443.80
50,000	\$4,148.20
500,000	\$41,192.20

Scenario 3 — Aggressive Protection (50% Reduction)

Users	CloudFront Cost
500	\$53
5,000	\$317
50,000	\$2,963
500,000	\$29,423

Monthly CloudFront Cost — 100 Video Library

Scenario 1 — No Bandwidth Protection

Users	CloudFront Cost
500	\$212
5,000	\$1,268
50,000	\$11,852
500,000	\$117,692

Scenario 2 — Signed URL Protection (30% Reduction)

Users	CloudFront Cost
500	\$148.40
5,000	\$887.60
50,000	\$8,296.40
500,000	\$82,384.40

Scenario 3 — Aggressive Protection (50% Reduction)

Users	CloudFront Cost
-------	-----------------

500	\$106
5,000	\$634
50,000	\$5,926
500,000	\$58,846

Observation

CloudFront bandwidth is the **largest cost driver in the entire platform**.

3. Amazon EC2 (Backend API)

Purpose

Amazon EC2 runs the **Node.js backend API**.

The backend handles:

- user authentication
 - course enrollment
 - learning progress tracking
 - API endpoints
 - video metadata retrieval
 - signed URL generation for video playback
-

High Availability Architecture

The platform uses an **Auto Scaling Group with two EC2 instances** behind a load balancer.

Benefits:

- no single point of failure
 - improved reliability
 - scalable API layer
-

Monthly Cost

Component	Cost
2 EC2 instances (t3.micro)	\$36

Observation

Backend compute cost remains nearly constant regardless of video streaming scale.

4. Application Load Balancer

Purpose

The Application Load Balancer distributes incoming API requests across multiple backend servers.

Benefits include:

- improved reliability
 - horizontal scaling
 - automatic failover
 - traffic distribution
-

Monthly Cost

Component	Cost
Application Load Balancer	\$18

5. MongoDB Database

Purpose

MongoDB stores all application data including:

- user accounts
- course metadata
- video metadata
- enrollment data
- progress tracking

The platform uses **MongoDB Atlas managed hosting**.

Monthly Cost

Component	Cost
MongoDB Atlas cluster	\$9

6. Amazon CloudWatch Monitoring

Purpose

CloudWatch monitors infrastructure health and application performance.

It provides:

- server metrics
 - API latency monitoring
 - error tracking
 - automated alerts
-

Monthly Cost

Component	Cost
CloudWatch monitoring	\$3

7. Amazon Route 53 DNS

Purpose

Route 53 manages the platform's domain and DNS routing.

Responsibilities include:

- routing users to CloudFront
 - routing API traffic to the load balancer
 - global DNS resolution
-

Monthly Cost

Component	Cost
Route 53	\$1

Combined Infrastructure Cost

After calculating each component individually, the total infrastructure cost can be determined.

Total Monthly Cost — 50 Video Library

Scenario 1 — No Bandwidth Reduction

Users	S3	CloudFront	EC2	ALB	MongoDB	CW	Route53	Total
500	\$3.46	\$106	\$36	\$18	\$9	\$3	\$1	\$176.46
5,000	\$34.56	\$634	\$36	\$18	\$9	\$3	\$1	\$735.56
50,000	\$345.60	\$5,926	\$36	\$18	\$9	\$3	\$1	\$6,338.60
500,000	\$3,456	\$58,846	\$36	\$18	\$9	\$3	\$1	\$62,369.00

Scenario 2 — 30% Bandwidth Reduction

Users	Total Monthly Cost
500	\$144.66
5,000	\$545.36
50,000	\$4,560.80
500,000	\$44,715.20

Scenario 3 — 50% Bandwidth Reduction

Users	Total Monthly Cost
500	\$123.46
5,000	\$418.56
50,000	\$3,375.60
500,000	\$32,946.00

Total Monthly Cost — 100 Video Library

Scenario 1 — No Bandwidth Reduction

Users	Total
500	\$285.92
5,000	\$1,404.12
50,000	\$12,610.20
500,000	\$124,671.00

Scenario 2 — 30% Bandwidth Reduction

Users	Total
500	\$222.32
5,000	\$1,023.72
50,000	\$9,054.60
500,000	\$89,363.40

Scenario 3 — 50% Bandwidth Reduction

Users	Total
500	\$179.92
5,000	\$770.12
50,000	\$6,684.20
500,000	\$65,825.00

Key Insights

1. Bandwidth Dominates Infrastructure Cost

More than **95% of total infrastructure cost** comes from CloudFront bandwidth.

2. Backend Infrastructure is Inexpensive

Backend services (EC2 + MongoDB + ALB + monitoring) together cost approximately:

~\$67 per month

This remains relatively stable until very large scale.

3. Storage Cost is Minimal

Even doubling the video library increases storage cost by only a few dollars per month.

4. Platform Scaling is Driven by Streaming Traffic

Example (50 video library):

Users	Monthly Cost
500	\$176
500,000	\$62,369

The platform cost scales primarily with **video streaming demand** rather than backend compute.

5. MongoDB Database (MongoDB Atlas)

Purpose

MongoDB serves as the primary database for the platform, storing all application data required for the MERN stack system to function.

In this architecture, the database stores:

- User accounts and authentication information
- Course and lesson metadata
- Video metadata (title, duration, resolution, S3 location)
- Course enrollment data
- Student progress tracking
- Learning history and analytics
- Platform configuration data

The platform uses MongoDB Atlas, which is a fully managed cloud database service for MongoDB.

Instead of manually installing and managing MongoDB servers on EC2, Atlas provides a managed environment that handles infrastructure, scaling, and maintenance automatically.

MongoDB Atlas Hosting Plan Used

The cost estimate assumes the MongoDB Atlas Shared Cluster (M2 tier).

Typical configuration of this plan:

Feature	Value
Cluster Tier	M2 Shared Cluster
Monthly Cost	~\$9/month
RAM	~2 GB shared
Storage	~2 GB included
Storage Type	SSD

Backup **Basic snapshot backups**

Deployment **Managed by MongoDB
Atlas**

This tier is designed for early-stage applications and small production workloads.

Why the Cost is \$9

The \$9 monthly cost represents the price of running a managed MongoDB database cluster in MongoDB Atlas.

This price includes several services that would otherwise require manual setup if running MongoDB on EC2.

1. Managed Database Infrastructure

MongoDB Atlas provides:

- pre-configured MongoDB servers
- automated setup
- database configuration management

Developers do not need to manage:

- database installation
- server configuration
- database upgrades
- security patching

This significantly reduces operational complexity.

2. Database Storage

The M2 cluster includes approximately:

2 GB of SSD database storage

This storage holds:

- user documents

- course data
- metadata
- indexing structures

Educational platforms typically store mostly metadata, not video files. Since videos are stored in Amazon S3, the database size remains relatively small.

Example database data:

Data Type	Typical Size
User profile	2–5 KB
Course metadata	10–20 KB
Video metadata	5–10 KB
Enrollment record	2–5 KB

Even with hundreds of thousands of records, the database often stays under a few gigabytes.

3. Managed Backups

MongoDB Atlas includes automated snapshot backups.

This ensures:

- database recovery in case of failure
- protection against accidental deletion
- rollback capability

Without Atlas, implementing backups would require:

- backup scripts
- storage infrastructure
- recovery procedures

4. Security and Access Control

The Atlas cluster also includes built-in security features:

- network access control
- IP allowlisting
- database user authentication
- TLS encrypted connections

These security features protect the platform's user data and prevent unauthorized access.

5. Automatic Maintenance

MongoDB Atlas performs several operational tasks automatically:

- database patching
- performance tuning
- monitoring
- infrastructure maintenance

This removes the need for a dedicated database administrator during early stages of the platform.

Why Database Cost Does Not Scale with Users (Initially)

Unlike video streaming bandwidth, the database cost does not increase significantly with user growth in the early stages.

Reason:

The platform stores small structured records, not large media files.

Example:

Users	Estimated DB Size
-------	----------------------

500 users ~20–50 MB

5,000 users ~200–400 MB

**50,000
users ~1–2 GB**

Since the M2 cluster includes 2 GB of storage, the database can support tens of thousands of users before needing an upgrade.

When MongoDB Costs Would Increase

The database would need to upgrade to larger clusters when:

- user base grows significantly
- database size exceeds storage limits
- query traffic increases

Typical upgrade path:

Atlas Tier	Monthly Cost	Use Case
M2	\$9	Small production apps
M5	~\$25	Growing startups
M10	~\$65	Medium production systems
M20+	\$150+	Large platforms

However, for the infrastructure scale discussed in this report (up to 50,000 users), the \$9 M2 cluster remains sufficient.